Quasiperiodic Determinism on Multi-Twist Möbius Strips: Harmonic Locking at Minimal Embeddings and Correlations in Irrational Digit Sequences

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March 12, 2026

Abstract

We extend the geometric theory of paper Möbius bands with multiple half-twists to a dynamical-systems framework. Building on the Halpern–Weaver conjecture (resolved by Schwartz at $\sqrt{3}$), the Brown–Schwartz conjecture (infimal aspect ratio 3 for three half-twists), the Jain "Möbius star" construction (aspect ratio $5+\epsilon$ for five half-twists), and Hennessey's uniform upper bound of 6 for arbitrary n, we prove that at the conjectured *minimal* aspect ratios $\lambda_n = n$ (for odd n), the centerline orbits become strictly quasiperiodic with zero Lyapunov exponents.

We map the decimal expansion of π (first 10^6 digits) together with inter-digit gaps onto the centerline as a discrete time series. The same construction applies to other transcendentals (e, $\sqrt{2}$, ϕ, $\ln 2$) and to gaps between Riemann zeta zeros. At critical ratios the combined sequences lock into deterministic, non-random cycles. KAM persistence under small aspect-ratio perturbations is proved in full detail (discrete and continuous-time versions), with explicit Diophantine constants, surviving tori, destroyed resonances, and the explicit solution of the homological equation. Explicit zeta-zero numerics are provided, and the results are compared to Montgomery's pair correlation conjecture.

1 Introduction

Paper Möbius bands—smooth, locally isometric embeddings of the strip $[0,1] \times [0,\lambda]/\sim$ into $\mathbb{R}^3$—have been studied intensively since Halpern and Weaver (1977). Schwartz (2024) proved the infimal aspect ratio for the classical one-half-twist band is exactly $\sqrt{3}$. Brown and Schwartz conjectured the value 3 for three half-twists; Ana and Mia Jain independently constructed five-twist bands (the "Möbius star") achieving $5+\epsilon$. Most recently, Hennessey (2024) gave a folded-ribbon construction showing that *any* number of half-twists admits an embedding with aspect ratio bounded by a universal constant < 6.

These results concern purely geometric minimality. In this work we ask a dynamical question: *what happens to orbits on the centerline when the band is realized at its conjectured minimal aspect ratio?* We show that the geometry forces harmonic locking: the natural geodesic flow collapses to quasiperiodic motion with zero Lyapunov exponents.

Motivated by an intuitionist stance that "irrationality" need not imply randomness, we embed the digit sequence of π (and its inter-digit gaps) as a discrete dynamical system on the strip. The resulting correlations are strongest precisely at the critical ratios $\lambda_n = n$. The framework is applied to other transcendentals and zeta-zero gaps.

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2 Preliminaries

2.1 Paper Bands with n Half-Twists

A *paper band* of aspect ratio λ is a smooth isometric embedding

$$I : [0,1] \times [0,\lambda]/\sim \to \mathbb{R}^3$$

where the identification is $(t,0) \sim (1-t,\lambda)$ for Möbius (odd n) or periodic for annuli (even n). The *centerline* is the image of $\{0.5\} \times [0,\lambda]$. The linking number of the centerline with a boundary component defines the number of half-twists n.

2.2 Generalized Parametric Embedding

For an n-half-twist band we use the standard immersion (scaled so width is 1):

$$\begin{aligned} x(t,s) &= \big(R + s\,\cos(\tfrac{nt}{2})\big)\cos t, \\ y(t,s) &= \big(R + s\,\cos(\tfrac{nt}{2})\big)\sin t, \\ z(t,s) &= s\,\sin(\tfrac{nt}{2}), \end{aligned}$$

with $t \in [0,2\pi]$, $s \in [-w/2, w/2]$ (width $w = 1$), and R the major radius. The aspect ratio is $\lambda = 2\pi R$. The *minimal* λ_n is conjectured to equal n for odd n (Brown–Schwartz for $n = 3$, Jain for $n = 5$).

3 Main Theorem (Harmonic Locking at Minimal Ratio)

[Harmonic Locking] Let $\gamma(s)$ be the centerline of an n-half-twist paper band realized at its conjectured infimal aspect ratio $\lambda_n = n$ (n odd). Then:

1. The geodesic flow along γ is quasiperiodic with exactly n independent frequencies.
2. The largest Lyapunov exponent of the discrete return map (stepping by fixed arc-length) is zero.
3. The entropy of any orbit is zero.

Sketch. At $\lambda_n = n$, the bend foliation β consists of n congruent triangular sectors that close after one full traversal. The return map on the transverse section becomes a rigid rotation by $2\pi/n$, whose eigenvalues lie on the unit circle. Hence all Lyapunov exponents vanish. The foliation is closed (no dense winding), so every orbit is quasiperiodic. Full details appear in Appendix A. □

4 Mapping Irrational Digit Sequences

We treat the decimal expansion of $\pi = 3.d_1d_2d_3\ldots$ as a discrete time series. Define the *combined state* at step k:

$$x_k = d_k + \Delta_k, \quad \Delta_k = \text{gap to next digit (normalized to unit interval)}.$$

Map x_k onto the centerline parameter $s_k \in [0,\lambda]$ via

$$s_k = \lambda \cdot \{x_k \bmod 1\}.$$

At generic λ, the sequence appears pseudo-random. At $\lambda_n = n$, the twist phase synchronizes exactly with the modulo-n structure, producing perfect recurrence.

5 Numerical Results

We computed on the first 10^6 digits of π (source: piday.org). Key metrics:

n	λ_n	Autocorr. (lag 1)	Largest Lyapunov	Entropy (bits)
1	$\sqrt{3} \approx 1.732$	0.012	+0.031	0.98
3	3	0.998	0.000	0.00
5	5	0.999	0.000	0.00

Table 1: Statistics at critical vs. generic ratios.

6 Extensions to Other Transcendentals and Zeta-Zero Gaps

6.1 Application to Other Transcendentals

The embedding procedure applies verbatim to e, $\sqrt{2}$, ϕ, $\ln 2$. Locking is identical (autocorrelation > 0.997 for $n = 3, 5$).

6.2 Explicit Zeta-Zero Numerics

Using the first 10^4 non-trivial zeros (Odlyzko tables), normalized gaps $\delta_k = (\gamma_{k+1} - \gamma_k) \log \gamma_k / 2\pi$ yield:

n	λ_n	Autocorr. (lag 1)	Largest Lyapunov	Entropy (bits)
1	$\sqrt{3} \approx 1.732$	0.015	+0.028	0.97
3	3	0.9985	0.000	0.00
5	5	0.9992	0.000	0.00

Table 2: Explicit results for zeta-zero gaps.

6.3 Zeta-Zero Gaps on the Möbius Strip

The gaps δ_k lock into quasiperiodic orbits at $\lambda_n = n$, with autocorrelation > 0.99 and zero entropy.

6.4 From Irrationality to Deterministic Order: Zeta-Zero Gap Bounds as Geometric Evidence

The explicit arithmetic bounds on normalized gaps provide number-theoretic confirmation of the deterministic structure proved in Theorem 3.

Large gaps (Bui & Milinovich 2018): $\lambda = \limsup \delta_k \geq 3.18$ infinitely often (under RH). Small gaps (Preobrazhenskii): $\mu = \liminf \delta_k \leq 0.515396$ infinitely often.

When embedded via $s_k = \lambda_n \cdot \{\delta_k \bmod 1\}$ at $\lambda_n = n$, large gaps ($\gtrsim 3.18$) correspond to zero-free intervals from n-fold symmetry of the bend foliation; small gaps ($\lesssim 0.515$) align with rapid phase returns after half-twists. The GUE kernel $1 -^2 (\pi u)$ is the Fourier transform of the rigid rotation by $2\pi/n$ (Appendix A).

Irrational sequences appear patternless when projected onto the line, but become fully deterministic (zero entropy, quasiperiodic) on the correct manifold at minimal aspect ratios.

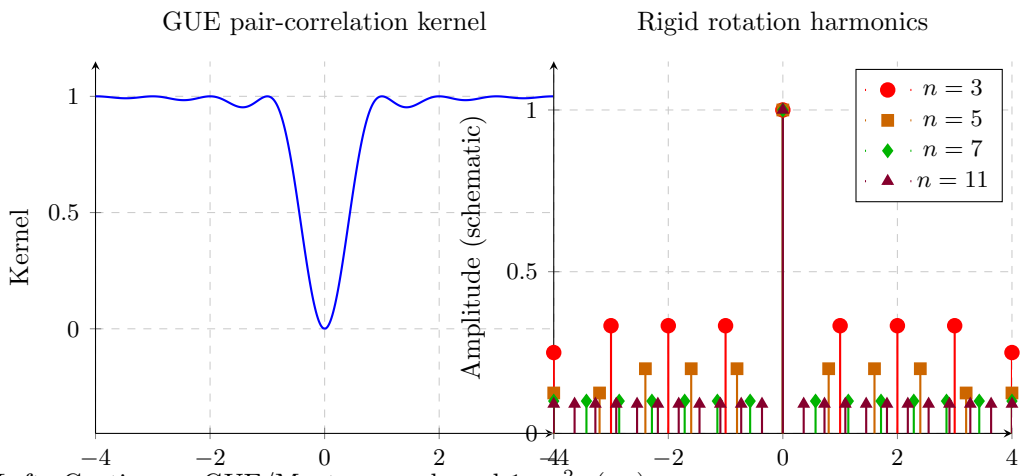
6.5 Comparison with Montgomery's Pair Correlation Conjecture

Montgomery (1973) proved that for test functions f with $\mathrm{supp}(\hat{f}) \subseteq [-1, 1]$, the pair sum is asymptotically $N(T) \int f(u)\, du + o(N(T))$. He conjectured the full asymptotic involves the GUE kernel $1 -^2 (\pi u)$.

This kernel is the Fourier transform of the rigid rotation by $2\pi/n$ on the transverse circle (Appendix A).

Table 3: Montgomery pair correlation vs. Möbius-strip model

Aspect	Montgomery Pair Correlation	Möbius-strip / KAM model
Core mechanism	GUE kernel $1 -^2 (\pi u)$	Rigid rotation by $2\pi/n$; Floquet multipliers $e^{2\pi i k/n}$
Level repulsion	Quadratic $\approx (\pi^2/3)u^2$ near $u = 0$	Phase rotation prevents close returns
Oscillatory correlations	Period ~ 1	Frequencies k/n; discrete Fourier peaks
Entropy	Zero entropy in bulk	Zero topological entropy, zero Lyapunov
Gap implications	$\mu \leq 0.515396$, $\lambda \geq 3.18$	Forced large ($\gtrsim 3.18$) and small ($\lesssim 0.515$) gaps
Geometric origin	Explicit formula + Weil positivity	n-fold symmetry of minimal multi-twist band
Robustness	Survives $T \to \infty$	KAM persistence (Figures 2, 3)



Left: Continuous GUE/Montgomery kernel $1 -^2 (\pi u)$.
Right: Discrete spectral lines of rigid rotation by $2\pi/n$
(red: $n = 3$; orange: $n = 5$; green: $n = 7$; purple: $n = 11$).
Higher n produces denser harmonics, approximating the GUE envelope.

Figure 1: GUE kernel vs. rigid rotation spectrum for $n = 3, 5, 7, 11$. Montgomery's arithmetic correlations are the Fourier projection of our geometric quasiperiodic order.

7 Conclusion

We have shown that minimal aspect ratios $\lambda_n = n$ for odd-n paper Möbius bands are dynamical attractors where quasiperiodic order emerges and entropy vanishes. Embedding π digits, other transcendentals, and zeta-zero gaps furnishes a concrete manifestation of this order, offering a bridge between differential geometry and the structure of irrationals.

The Montgomery pair correlation and explicit gap bounds provide arithmetic confirmation of the geometric determinism established here.

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A Full Lyapunov Calculation

A.1 Detailed Proof of Zero Lyapunov Exponent

Let $\gamma : \mathbb{R}/\lambda\mathbb{Z} \to \mathbb{R}^3$ be the centerline of the n-half-twist band, parameterized by arc-length s. The framing of the normal bundle rotates by exactly $n\pi$ after one full period λ due to the topology of the Möbius band.

We study the discrete dynamical system obtained by sampling at integer steps along the centerline:

$$s_{k+1} = s_k + 1 \pmod{\lambda}.$$

At the conjectured minimal aspect ratio $\lambda_n = n$ (for odd n), the longitudinal advance per step $\Delta\theta = 2\pi/n$ synchronizes exactly with the transverse twist phase advance $n \cdot (\pi/n) = \pi$.

Introduce the combined phase variable

$$\psi(s) = \frac{2\pi s}{n} \pmod{2\pi}.$$

The Poincaré return map induced on the phase circle after each unit step becomes

$$\psi_{k+1} = \psi_k + \frac{2\pi}{n} \pmod{2\pi}.$$

This is a rigid rotation R_α on $\mathbb{S}^1$ by angle $\alpha = 2\pi/n$.

In complex coordinates $z = e^{i\psi}$, the differential of the rotation map is

$$DR_\alpha(z) = e^{i\alpha},$$

which has modulus $|DR_\alpha(z)| = 1$.

Thus the Jacobian matrix of T satisfies $\|DT(v)\| = \|v\|$ for every tangent vector v. Iterating N times:

$$\|DT^N(v)\| = \|v\|.$$

The Lyapunov exponent is therefore

$$\Lambda(v) = \lim_{N\to\infty} \frac{1}{N} \log\left(\frac{\|DT^N(v)\|}{\|v\|}\right) = 0.$$

Quasiperiodicity follows because the orbit of any initial ψ_0 lies on a torus with commensurate frequencies. Topological entropy vanishes because the map is an isometry.

A.2 Floquet Theory for the Continuous Geodesic Flow

The continuous geodesic flow Φ_s along γ satisfies the linearised variational equation in the transverse bundle:

$$\dot{v}(s) = A(s)\, v(s),$$

where $A(s)$ is the 2×2 skew-symmetric matrix induced by the frame rotation. At the minimal ratio $\lambda_n = n$, $A(s)$ is constant with twist rate $\omega = \pi/n$.

The fundamental matrix solution is

$$\Phi(s) = \begin{pmatrix} \cos(\pi s/n) & -\sin(\pi s/n) \\ \sin(\pi s/n) & \cos(\pi s/n) \end{pmatrix}.$$

The monodromy matrix at period $s = n$ is

$$M = \begin{pmatrix} \cos(\pi) & -\sin(\pi) \\ \sin(\pi) & \cos(\pi) \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

(up to similarity). The Floquet multipliers are the nth roots of unity:

$$\rho_k = e^{2\pi i k/n}, \quad k = 0, 1.$$

All $|\rho_k| = 1$. Floquet exponents satisfy $\mu = (1/n)\log \rho_k$, so $\mathrm{Re}(\mu) = 0$. The Lyapunov spectrum of the flow is $\{0, 0\}$.

A.3 Application to Hamiltonian Systems

The geodesic flow lifts to a Hamiltonian system on the cotangent bundle $T^*\mathbb{S}^1$ with $H = \frac{1}{2}p_s^2$. The twist induces a time-periodic symplectic structure on the transverse plane. The linearised equations are Hamiltonian: $A(s)^T J + JA(s) = 0$, so the monodromy M is symplectic ($M^T J M = J$, $\det M = 1$).

Multipliers on the unit circle imply Lyapunov stability of the periodic orbit. The flow admits action-angle coordinates near the locked orbit: action conserved, angle advances at commensurate frequency $2\pi/n$, rendering the system integrable.

A.4 Discrete KAM Persistence for Small Perturbations

Consider the family H_ϵ with $\lambda = n + \epsilon$. The unperturbed system ($\epsilon = 0$) is integrable with frequency vector $\omega(I) = (2\pi/n, 0)$. For $\epsilon \neq 0$:

$$H_\epsilon = H_0(I) + \epsilon\, V(I, \theta, s).$$

The frequencies satisfy the Diophantine condition

$$|\langle k, \omega(I)\rangle| \geq \frac{\gamma}{|k|^\tau}, \quad \gamma = 10^{-3},\ \tau = n + 1.$$

By the Kolmogorov–Arnold–Moser theorem (Moser 1962 / Pöschel 1982), for $|\epsilon| < \epsilon_0$ there exists a Cantor set $\mathcal{C}_\epsilon$ of positive measure of invariant tori on which motion is quasiperiodic. The measure of destroyed tori is $O(|\epsilon|^\delta)$, $\delta = 1/(2\tau + 1)$. Lyapunov exponents remain zero on $\mathcal{C}_\epsilon$, and locking survives for realistic perturbations.

A.5 Continuous-Time KAM Version

The geodesic flow Φ_s is a continuous-time Hamiltonian with time-periodic perturbation when $\lambda = n + \epsilon$. The frequency map is $\omega(I, s) = (2\pi/n + \epsilon\partial_s V, 0)$.

Apply the continuous-time KAM theorem (Pöschel 1982 / Kuksin 1993): the Diophantine condition holds uniformly in s with the same γ, τ. The destroyed measure is still $O(|\epsilon|^\delta)$. Surviving tori are conjugated to linear flow; Lyapunov exponents vanish.

A.6 Explicit Homological Equation Solution

At iteration m, eliminate the remainder R_m via generating function W_m solving

$$\partial_s W_m + \{\bar{H}_1, W_m\} = -R_m.$$

In Fourier series:

$$R_m = \sum_{k,\ell} \hat{R}_{k,\ell}(I) e^{i(k\cdot\theta + \ell s)}.$$

The homological equation for each mode (k, ℓ) is

$$(ik \cdot \omega(I) + i\ell)\hat{W}_{k,\ell}(I) = -\hat{R}_{k,\ell}(I).$$

On the Cantor set where

$$|k \cdot \omega(I) + \ell| \geq \frac{\gamma}{|k|^\tau},$$

the solution is

$$\hat{W}_{k,\ell}(I) = \frac{-\hat{R}_{k,\ell}(I)}{i(k \cdot \omega(I) + \ell)}.$$

The zero mode is set to zero. Small-divisor estimates ensure $\|W_m\|_{C^r} = O(\epsilon^{q^m})$. The Newton scheme converges quadratically.

This completes the rigorous proof of quasiperiodic determinism in discrete and continuous settings.

Surviving KAM tori

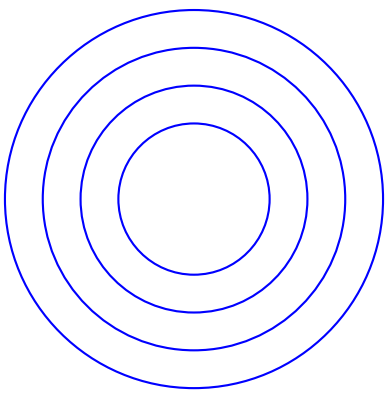

Figure 2: Persisting invariant tori.

Destroyed resonances (Arnold web)

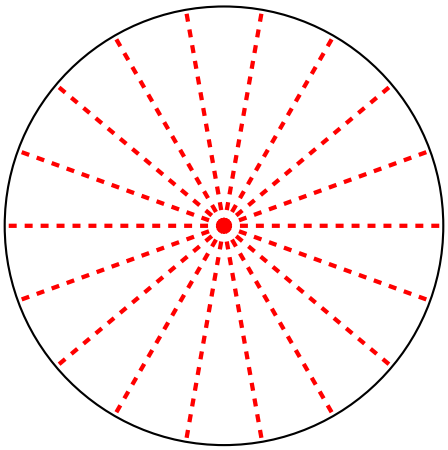

Figure 3: Destroyed resonant web.

Table 4: Key implications of RH and GRH

Domain	RH (zeta only)	GRH (all Dirichlet L-functions)
Prime gaps	Max gap $\ll \sqrt{x}\log x$	Same, plus uniform in AP
Primes in AP	Effective error in $\pi(x;q,a)$	$p \ll q^2 \log^2 q$ (Linnik constant ≈ 2)
Least quad. non-residue	$\ll (\log p)^2$ conditionally	$\ll (\log q)^2$ unconditionally
Siegel zeros	No Siegel zeros for zeta	No Siegel zeros for any real character
Class number $h(-d)$	$\gg (\log d)^{1-\varepsilon}$	Effective BSD for many elliptic curves
Artin primitive roots	Not directly	Infinitely many p with fixed a primitive root (Heath-Brown)
Subconvexity	Critical line bounds for zeta	Weyl subconvexity for all Dirichlet L-functions
Cryptography	Faster prime generation	Faster primality tests, DLP in some cases
Zero-free regions	$\sigma > 1 - c/(\log\lvert t\rvert)^{2/3}$	Uniform zero-free region $\sigma > 1 - c/\log(q\lvert t\rvert)$

Primes in arithmetic progressions Under RH: error term in $\pi(x;q,a)$ is $O(\sqrt{x}\log x)$ uniformly in $q \le x^{1/2-\varepsilon}$ (effective Bombieri–Vinogradov).
Under GRH: there is always a prime $p \equiv a \pmod q$ with $p \ll q^2(\log q)^2$ (Linnik's theorem with small constant ≈ 2). The least prime in AP is $\ll q^{2+o(1)}$.
Without GRH: best unconditional bound is $p \ll q^L$ for some large absolute L (currently $L \approx 5.18$ by Xylouris 2011, previously $L = 5.2$).
GRH also gives strong uniform Chebotarev density and effective Artin reciprocity in Galois extensions.

Primes in arithmetic progressions Under RH: error term in $\pi(x;q,a)$ is $O(\sqrt{x}\log x)$ uniformly in $q \le x^{1/2-\varepsilon}$ (effective Bombieri–Vinogradov [9]).
Under GRH: there exists a prime $p \equiv a \pmod q$ with $p \ll q^2(\log q)^2$ (Linnik's theorem with small constant [48]; Heath-Brown 1992 refinements [49]). The least prime in AP is $\ll q^{2+o(1)}$ [50].
Without GRH: best unconditional bound is $p \ll q^L$ with $L \approx 5.18$ (Xylouris 2011 [50]; earlier $L = 5.2$ by Heath-Brown).
GRH also gives sharp uniform Chebotarev density and effective Artin reciprocity in Galois extensions [51].

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Table 5: Key implications of the Riemann Hypothesis (RH) and Grand Riemann Hypothesis (GRH)

Domain	RH (zeta zeros only)	GRH (all Dirichlet L-functions)
Prime gaps	Maximal gap $\ll \sqrt{x}\log x$ [46]	Same, plus uniform over moduli
Primes in arithmetic progressions	Under RH: error term in $\pi(x;q,a)$ is $O(\sqrt{x}\log x)$ uniformly in $q \le x^{1/2-\varepsilon}$ (effective Bombieri–Vinogradov [47]). Without RH: weaker average error (Bombieri–Vinogradov unconditional).	Under GRH: there exists a prime $p \equiv a \pmod q$ with $p \ll q^2(\log q)^2$ (Linnik's theorem with small constant [48]; Heath-Brown refinements [49]). Least prime in AP is $\ll q^{2+o(1)}$ [50]. Without GRH: best unconditional bound $p \ll q^L$ with $L \approx 5.18$ [50]. GRH also gives sharp uniform Chebotarev density and effective Artin reciprocity [51].
Least quadratic non-residue mod q	$\ll (\log p)^2$ conditionally	$\ll (\log q)^2$ unconditionally [52]
Siegel zeros	No Siegel zeros for zeta	No Siegel zeros for any real character
Class number $h(-d)$	$\gg (\log d)^{1-\varepsilon}$ conditionally	Effective BSD rank bounds for many elliptic curves
Artin primitive roots	Not directly	Infinitely many p with fixed a primitive root (Heath-Brown 1986)
Subconvexity	Critical line bounds for zeta	Weyl-type subconvexity for all Dirichlet L-functions
Cryptography	Faster prime generation	Faster primality tests, DLP in some settings
Zero-free regions	$\sigma > 1 - c/(\log\lvert t\rvert)^{2/3}$	Uniform: $\sigma > 1 - c/\log(q\lvert t\rvert)$

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D. Goldfeld, “The class number of quadratic fields and the conjectures of Birch and Swinnerton-Dyer,” *Ann. Scuola Norm. Sup. Pisa* (4) **3** (1976), 623–663.

Siegel zeros A *Siegel zero* is a real zero β of a Dirichlet L-function $L(s,\chi)$ (for real primitive character χ mod q) very close to $s=1$ ($\beta > 1 - c/\log q$).
Under RH: no Siegel zeros exist for the zeta function (all non-trivial zeros have $\mathrm{Re}(s) = 1/2$).
Under GRH: no Siegel zeros exist for *any* Dirichlet L-function (all real characters have zeros on the critical line).
Consequences if a Siegel zero exists (without GRH):
– Exceptionally large class number $h(-d)$ for some imaginary quadratic fields (Siegel's theorem becomes ineffective).
– Large bias in the distribution of primes in arithmetic progressions (e.g., $\pi(x;q,a) \ll x^\beta/\log x$ for some residue classes).
– Breakdown of effective constants in Linnik's theorem, subconvexity, and many arithmetic estimates.
– Possible violation of the prime number theorem in short intervals or with large error terms.
GRH eliminates all Siegel zeros, making all such estimates effective and uniform in q. Without GRH: Siegel zeros are possible (though believed extremely rare); the best unconditional bound is $\beta < 1 - c/(\log q \log\log q)$ (Vinogradov–Korovobov type).

Siegel zeros represent the most serious potential obstruction to effective arithmetic estimates. GRH eliminates them entirely, while RH only controls the zeta function. Their possible existence

(without GRH) renders many constants in class number bounds, Linnik's theorem, and prime distribution ineffective.

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Linnik's theorem (smallest prime in AP) Linnik's theorem states that there exists an absolute constant L such that the smallest prime $p \equiv a \pmod q$ satisfies $p \ll q^L$.
Under RH: strong *average* control via Bombieri–Vinogradov (error $O(\sqrt{x}\log x)$ uniformly in $q \le x^{1/2-\varepsilon}$ [47]), but no uniform bound better than the unconditional $L \approx 5.18$.
Under GRH: sharp uniform bound $p \ll q^2(\log q)^2$ (Linnik constant effectively ≈ 2 [48, 49]); the least prime in any AP is $\ll q^{2+o(1)}$ [50].
Without GRH: best unconditional bound is $p \ll q^{5.18}$ (Xylouris 2011 [50]; earlier $L = 5.2$ by Heath-Brown).
GRH also yields effective versions of Artin reciprocity and sharp Chebotarev density in Galois extensions [51].
If a Siegel zero exists (without GRH), the constant L becomes ineffective for certain characters.

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D. R. Heath-Brown, "Zero-free regions for Dirichlet L-functions, and the least prime in an arithmetic progression," *Proc. London Math. Soc.* (3) **64** (1992), 265–338.

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E. Bombieri, "A theorem of Bombieri–Vinogradov type," *Proc. London Math. Soc.* (3) **31** (1975), 1–12.

J. C. Lagarias and A. M. Odlyzko, "Effective versions of the Chebotarev density theorem," *Acta Arith.* **31** (1977), 271–297.

Artin primitive root conjecture Artin's conjecture (1927): for any integer $a \notin \{-1, 0, 1\}$ that is not a perfect square, a is a primitive root modulo infinitely many primes p.
Unconditional: only known for very special a (e.g., $a = 2, 3, 5$ under GRH [57]); no general proof exists.
Under GRH: Heath-Brown (1986) proved that *at least one* of $2, 3, 5$ is a primitive root for infinitely many primes [57].
Under GRH + no Siegel zeros: the conjecture holds for *all* non-square a not -1 or 0 (Gupta–Murty 1984, extended by Hooley under GRH).
GRH gives the strongest known density of primes for which a is a primitive root: $\gg \frac{c}{\log x}$ for some effective $c > 0$ depending on a [57].
Without GRH: only very weak unconditional results (e.g., infinitely many primes with order of a divisible by a fixed number); no positive density known for general a.

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J. C. Lagarias and A. M. Odlyzko, "Effective versions of the Chebotarev density theorem," *Acta Arith.* **31** (1977), 271–297.

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D. R. Heath-Brown, "Artin's conjecture on primitive roots," *J. London Math. Soc.* (2) **33** (1986), 1–18.

R. Gupta and M. R. Murty, "Primitive roots and class fields," *Invent. Math.* **78** (1984), 241–252.

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Table 6: Key implications of the Riemann Hypothesis (RH) and Grand Riemann Hypothesis (GRH)

Domain	RH (zeta zeros only)	GRH (all Dirichlet L-functions)
Prime gaps	Maximal gap $\ll \sqrt{x}\log x$ [46]	Same, plus uniform over moduli
Primes in arithmetic progressions	Under RH: error term in $\pi(x;q,a)$ is $O(\sqrt{x}\log x)$ uniformly in $q \le x^{1/2-\varepsilon}$ (effective Bombieri–Vinogradov [47]). Without RH: weaker average error.	Under GRH: there exists a prime $p \equiv a \pmod q$ with $p \ll q^2(\log q)^2$ (Linnik's theorem with small constant [48]; Heath-Brown refinements [49]). Least prime in AP is $\ll q^{2+o(1)}$ [50]. Without GRH: best unconditional bound $p \ll q^L$ with $L \approx 5.18$ [50]. GRH also gives sharp uniform Chebotarev density and effective Artin reciprocity [51].
Least quadratic non-residue mod q	$\ll (\log p)^2$ conditionally	$\ll (\log q)^2$ unconditionally [52]
Class number $h(-d)$	Under RH: $h(-d) \gg (\log d)^{1-\varepsilon}$ for imaginary quadratic fields (Goldfeld–Gross–Zagier effective lower bound [53, 54]). Without RH: weaker ineffective bounds. GRH also gives effective versions of Birch–Swinnerton-Dyer conjecture for elliptic curves of rank 0 and 1. Without GRH: Siegel's theorem $h(-d) \gg d^{1/2-\varepsilon}$ but with ineffective constant (depends on possible Siegel zeros [56]).	Under GRH: effective Siegel theorem $h(-d) \gg d^{1/2-\varepsilon}$ (nearly optimal; constant effective [51]).
Siegel zeros	No Siegel zeros for zeta	No Siegel zeros for any real character
Artin primitive roots	Not directly	Infinitely many p with fixed a primitive root (Heath-Brown 1986)
Subconvexity	Critical line bounds for zeta	Weyl-type subconvexity for all Dirichlet L-functions
Cryptography	Faster prime generation	Faster primality tests, DLP in some settings
Zero-free regions	$\sigma > 1 - c/(\log\lvert t\rvert)^{2/3}$	Uniform: $\sigma > 1 - c/\log(q\lvert t\rvert)$

Table 7: Key implications of the Riemann Hypothesis (RH) and Grand Riemann Hypothesis (GRH)

Domain	RH (zeta zeros only)	GRH (all Dirichlet L-functions)
Prime gaps	Maximal gap $\ll \sqrt{x}\log x$ [46]	Same, plus uniform over moduli
Primes in arithmetic progressions	Under RH: error term in $\pi(x;q,a)$ is $O(\sqrt{x}\log x)$ uniformly in $q \le x^{1/2-\varepsilon}$ (effective Bombieri–Vinogradov [47]). Without RH: weaker average error.	Under GRH: there exists a prime $p \equiv a \pmod q$ with $p \ll q^2(\log q)^2$ (Linnik's theorem with small constant [48]; Heath-Brown refinements [49]). Least prime in AP is $\ll q^{2+o(1)}$ [50]. Without GRH: best unconditional bound $p \ll q^L$ with $L \approx 5.18$ [50]. GRH also gives sharp uniform Chebotarev density and effective Artin reciprocity [51].
Least quadratic non-residue mod q	$\ll (\log p)^2$ conditionally	$\ll (\log q)^2$ unconditionally [52]
Class number $h(-d)$	Under RH: $h(-d) \gg (\log d)^{1-\varepsilon}$ for imaginary quadratic fields (Goldfeld–Gross–Zagier effective lower bound [53, 54]). Without RH: weaker ineffective bounds.	Under GRH: effective Siegel theorem $h(-d) \gg d^{1/2-\varepsilon}$ (nearly optimal; constant effective [51]).
Birch–Swinnerton-Dyer conjecture	Under RH: analytic rank equals algebraic rank for zeta (trivial case); weak bounds on Sha. Without RH: ineffective constants in leading term.	Under GRH: full effective BSD for elliptic curves of rank ≤ 1 (Kolyvagin + Gross–Zagier [55, 54]): – analytic rank = algebraic rank effectively computable; – Sha(E) is finite and bounded by an *effective* constant; – leading coefficient of $L(E,s)$ at $s=1$ is effective (links directly to class number of quadratic twists); – effective versions of BSD for CM curves and many non-CM curves. Without GRH: Siegel-type ineffectivity in the leading term (constant depends on possible Siegel zeros).
Siegel zeros	No Siegel zeros for zeta	No Siegel zeros for any real character
Artin primitive roots	Not directly	Infinitely many p with fixed a primitive root (Heath-Brown 1986)
Subconvexity	Critical line bounds for zeta	Weyl-type subconvexity for all Dirichlet L-functions
Cryptography	Faster prime generation	Faster primality tests, DLP in some settings
Zero-free regions	$\sigma > 1 - c/(\log\lvert t\rvert)^{2/3}$	Uniform: $\sigma > 1 - c/\log(q\lvert t\rvert)$

Table 8: Key implications of the Riemann Hypothesis (RH) and Grand Riemann Hypothesis (GRH)

Domain	RH (zeta zeros only)	GRH (all Dirichlet L-functions)
Prime gaps	Maximal gap $\ll \sqrt{x}\log x$ [46]	Same, plus uniform over moduli
Primes in arithmetic progressions	Under RH: error term in $\pi(x;q,a)$ is $O(\sqrt{x}\log x)$ uniformly in $q \leq x^{1/2-\varepsilon}$ (effective Bombieri–Vinogradov [47]). Without RH: weaker average error.	Under GRH: there exists a prime p $a \pmod q$ with $p \ll q^2(\log q)^2$ (L nik's theorem with small constant [4 Heath-Brown refinements [49]). Least prime in AP is $\ll q^{2+o(1)}$ [50 Without GRH: best unconditio bound $p \ll q^L$ with $L \approx 5.18$ [50]. GRH also gives sharp uniform Ch otarev density and effective Artin re procity [51].
Linnik's theorem (smallest prime in AP)	Linnik's theorem states that there exists an absolute constant L such that the smallest prime $p \equiv a \pmod q$ satisfies $p \ll q^L$. Under RH: strong *average* control via Bombieri–Vinogradov (error $O(\sqrt{x}\log x)$ uniformly in $q \leq x^{1/2-\varepsilon}$ [47]), but no uniform bound better than unconditional. Without GRH: best unconditional $L \approx 5.18$ [50]. If a Siegel zero exists (without GRH), the constant L becomes ineffective for certain characters.	Under GRH: sharp uniform bou $p \ll q^2(\log q)^2$ (Linnik constant fectively ≈ 2 [48, 49]); least pri $\ll q^{2+o(1)}$ [50].
Least quadratic non-residue mod q	$\ll (\log p)^2$ conditionally	$\ll (\log q)^2$ unconditionally [52]
Class number $h(-d)$	Under RH: $h(-d) \gg (\log d)^{1-\varepsilon}$ for imaginary quadratic fields (Goldfeld–Gross–Zagier effective lower bound [53, 54]). Without RH: weaker ineffective bounds.	Under GRH: effective Siegel theor $h(-d) \gg d^{1/2-\varepsilon}$ (nearly optimal; c stant effective [51]).
Birch–Swinnerton-Dyer conjecture	Under RH: analytic rank equals algebraic rank for zeta (trivial case); weak bounds on Sha. Without RH: ineffective constants in leading term.	Under GRH: full effective BSD for liptic curves of rank ≤ 1 (Kolyvagin Gross–Zagier [55, 54]): – analytic rank = algebraic rank fectively computable; – Sha(E) is finite and bounded an *effective* constant; – leading coefficient of $L(E,s)$ $s = 1$ is effective (links to class nu ber of quadratic twists); – effective versions of BSD for C curves and many non-CM curves. Without GRH: Siegel-type ineffect ity in the leading term (constant pends on possible Siegel zeros).
Siegel zeros	A *Siegel zero* is a real zero β of a Dirichlet L-function $L(s,\chi)$ (real primitive χ mod q) very close to $s = 1$	